MODEL ANSWERS

Answer 1:
Prokaryotic cells have no membrane-bound nucleus (DNA lies in a nucleoid) while eukaryotic cells have a true nucleus; prokaryotes lack membrane-bound organelles such as mitochondria and ER; prokaryotes have smaller 70S ribosomes, eukaryotes 80S. (Also: circular DNA vs linear chromosomes; prokaryotes are usually much smaller.)

Answer 2:
Mitochondria are the site of aerobic respiration, where the Krebs cycle and oxidative phosphorylation produce ATP. Muscle contraction uses ATP rapidly, so muscle cells have many mitochondria to supply enough ATP.

Answer 3:
An enzyme's active site has a shape complementary to its substrate. The substrate binds to form an enzyme-substrate complex; this lowers the activation energy so the reaction proceeds faster, and the enzyme is released unchanged. Raising temperature increases kinetic energy and collisions, so rate rises to an optimum (~37 deg C in humans); beyond it, bonds maintaining the tertiary structure break, the active site changes shape and the enzyme denatures. Each enzyme has an optimum pH (pepsin ~2, salivary amylase ~7); other pH values change the charges on R-groups, disrupting ionic and hydrogen bonds so the active site no longer fits the substrate.